figures-referenced

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
library(ggplot2)
library(patchwork)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

metadata <- read_csv(here("data", "metadata.csv"))

site_order <- c("tc_isco", "tc_1", "tc_15", "tc_2", "tc_3",
```

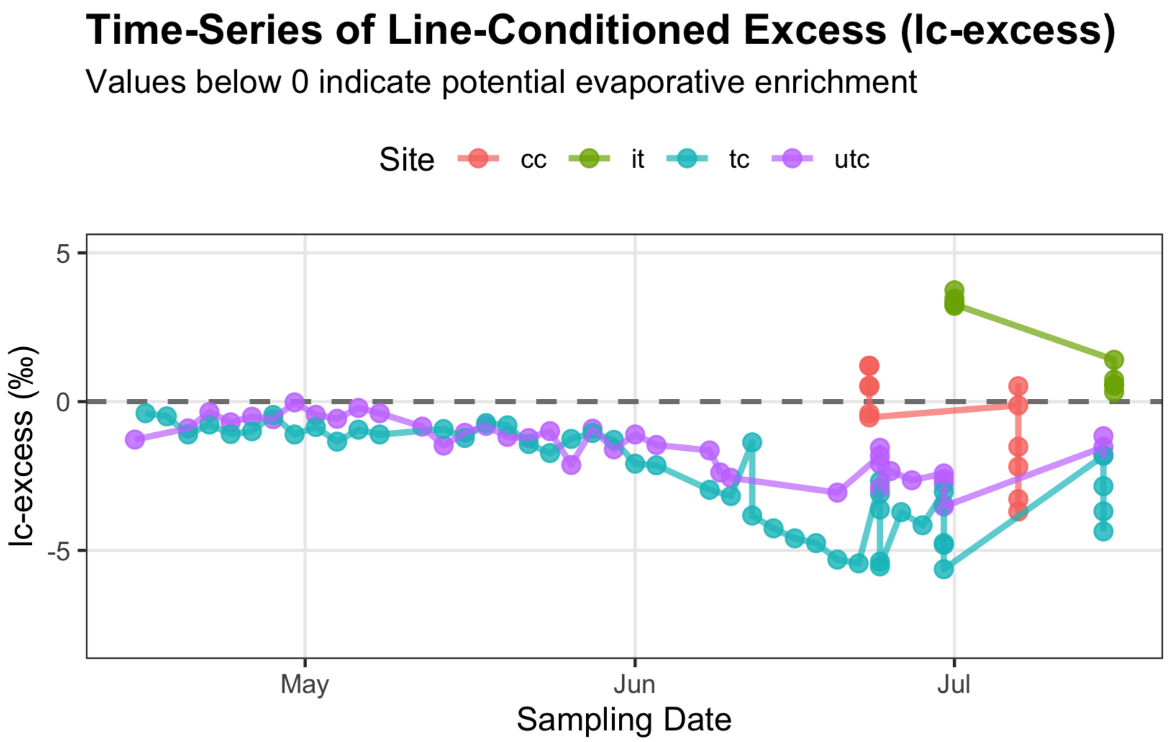
```r
                 "utc_isco", "tc_ds", "tc_mid", "tc_us")

trail_creek <- compiled_isotope |>
  filter(location %in% c("utc", "tc"))

coal_creek <- compiled_isotope |>
  filter(location %in% c("cc")) |>
  filter(site != "coal_3")

coal_creek_all <- compiled_isotope |>
 filter(location %in% c("cc"))

italian_creek <- compiled_isotope |>
  filter(location == "it")


# Downstream Trail Creek
tc_data <- read_csv(here("data", "TrailCreek_2026-07-23-15.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
         date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_data <- read_csv(here("data", "UTC_2026-07-23-15.csv"), show_col_types = FALSE) |> # read
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt,  # renaming columns to be shorter
         dp = differential_pressure_k_pa,
```

```
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
         date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE),  # averaging the data based on the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "utc") # creating a new column of the site

tc_utc_data <- merge(tc_data, utc_data, all = TRUE) # merging the tc and utc data

isco_data <- merge(
  x = tc_utc_data,
  y = compiled_isotope,
  by.x = c("site", "date"), # columns in tc_utc_data
  by.y = c("location", "date") # matching columns in compiled_isotope
)

tc_isco <- isco_data |>
  filter(site %in% c("tc"))

utc_isco <- isco_data |>
  filter(site %in% c("utc"))
```

Figure 1: Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = compiled_isotope,
       aes(x = date,
           y = lc_excess,
           color = location,
           group = location)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
```

```
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(linewidth = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (\u2030)",
       color = "Site") +
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```




```
# Buildings a look up table of stream distances
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
  mutate(xmax = dist_m,
```

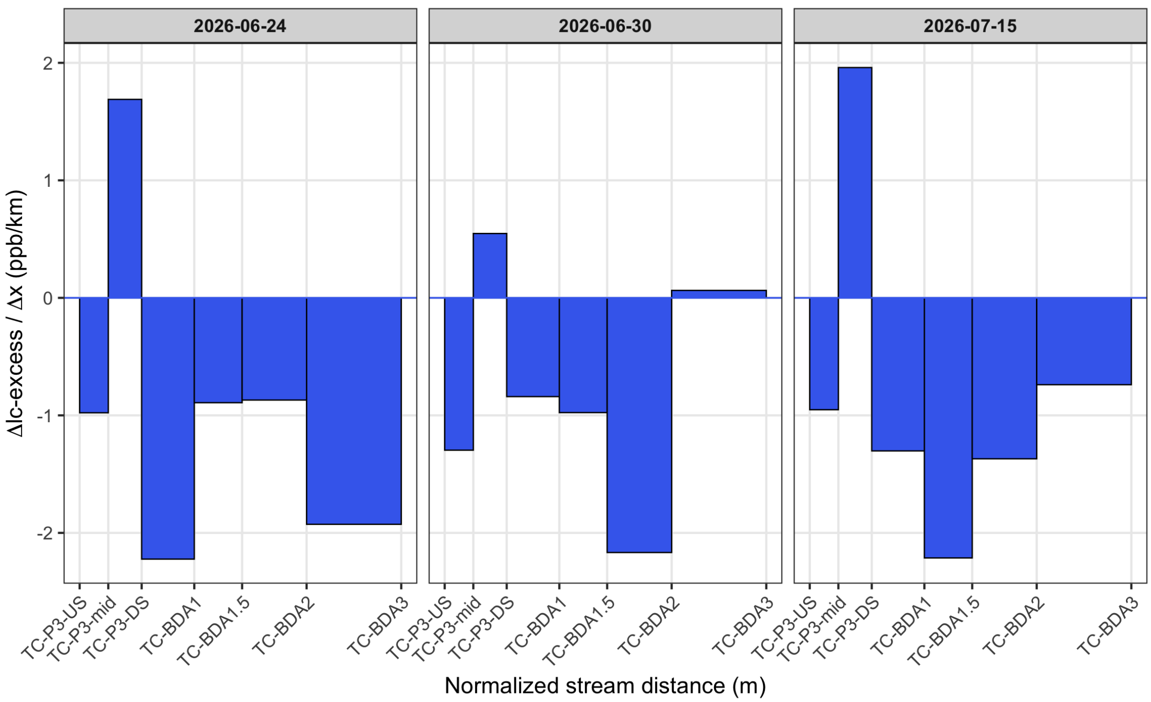
```
        xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first

# Creating a filtered data frame for lc-excess
gradient_lc_clean <- gradient_tc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δl
  pivot_longer( cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-excess/Δx_7-15`),
                names_to = "date",
                values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                          date == "Δlc-excess/Δx_6-30" ~ "2026-06-30",
                          date == "Δlc-excess/Δx_7-15" ~ "2026-07-15"),
        gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

# Gradient plot
plot1 <- ggplot(gradient_lc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site) +
  labs(title = "Trail Creek: Spatial variations in Δlc-excess/Δx across sampling dates",
       x = "Normalized stream distance (m)",
       y = expression(paste(Delta, "lc-excess / ", Delta, "x (ppb/km)"))) +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1))
# Buildings a look up table of stream distances

distances_cc <- tibble(
  site = c("CC1", "CC2", "CC4", "CC5", "CC6"),
  dist_m = c(0, 1496, 2006, 2478, 2885)) |>
  mutate(xmax = dist_m,
        xmin = lag(dist_m, default = 0))

# Creating a filtered data frame for lc-excess
```

```
gradient_cc_clean <- gradient_cc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`) |>
  pivot_longer(cols = c(`Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`),
    names_to = "date",
    values_to = "gradient") |>
  mutate(date = case_when(
    date == "Δlc-excess/Δx_6-23" ~ "2026-06-23",
    date == "Δlc-excess/Δx_7-7" ~ "2026-07-07"),
    gradient = as.numeric(gradient)) |>
  left_join(distances_cc |> select(site, xmin, xmax),
            by = "site") |>
  mutate(site = factor(site, levels = distances_cc$site)) |>
  filter(!is.na(gradient))


# Gradient plot
plot2 <- ggplot(gradient_cc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_cc$dist_m,
    labels = distances_cc$site) +
  labs(
    title = "Coal Creek: Spatial variations in Δlc-excess/Δx across sampling dates",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x (ppb/km)"))) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1))
```

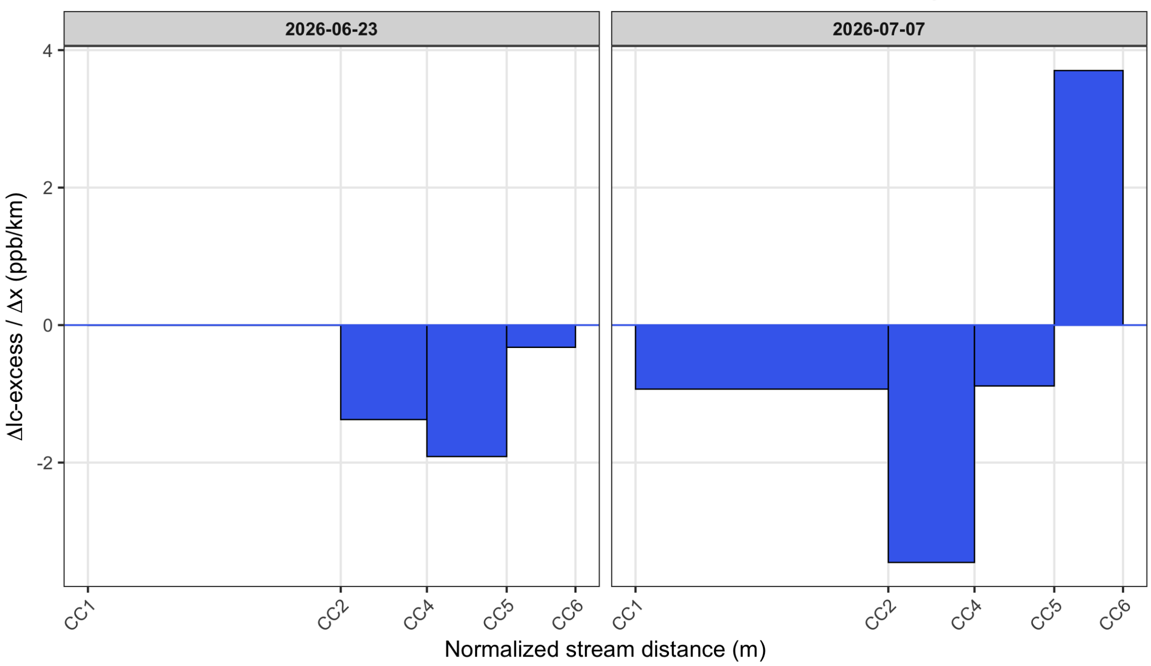
Figure 2:

```
combined_stacked <- plot1 / plot2

combined_stacked # print the graph
```

Trail Creek: Spatial variations in ΔIc-excess/Δx across sampling dates



Coal Creek: Spatial variations in ΔIc-excess/Δx across sampling dates



```
# averaging the lc-excess and including standard error
delta_summary <- compiled_isotope |>
  mutate(site = factor(site, levels = site_order)) |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

# creating a plot to show the mean lc-excess of each site (trail creek)
plot3 <- ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
              width = 0.3,
              color = "black") +
  scale_x_discrete(limits = c("tc_isco", "tc_1", "tc_15", "tc_2", "tc_3", "utc_isco", "tc_ds"
  labels = c("tc_isco" = "TC-ISCO", "tc_1" = "TC-1", "tc_15" = "TC-1.5", "tc_2" = "TC-2",
            "tc_3" = "TC-3", "tc_ds" = "IT-DS", "tc_mid" = "TC-Mid", "tc_us" = "TC-US",
            "utc_isco" = "UTC-ISCO")) +
  scale_fill_viridis_d(option ="G") + # color palette
  theme_minimal() +
  labs(title = "Trail Creek",
      x = "Sampling Site",
      y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1), # tilt labels
        legend.position = "none")

# averaging the lc-excess and including standard error
delta_sum_cc <- coal_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )


# creating a plot to show the mean lc-excess of each site (coal creek)
plot4 <- ggplot(delta_sum_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
```

```r
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
                width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G") +
  theme_minimal() +
  labs(title = "Coal Creek",
       x = "Sampling Site",
       y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")

# averaging the lc-excess and including standard error
delta_summary_it <- italian_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )


# creating a plot to show the mean lc-excess of each site (italian creek)
plot5 <- ggplot(delta_summary_it, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
                width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G") +
  theme_minimal() +
  labs(title = "Italian Creek",
       x = "Sampling Site",
       y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```

Figure 3:

```r
combined_stacked_2 <- plot3 + plot4 + plot5

combined_stacked_2 # print the graph
```

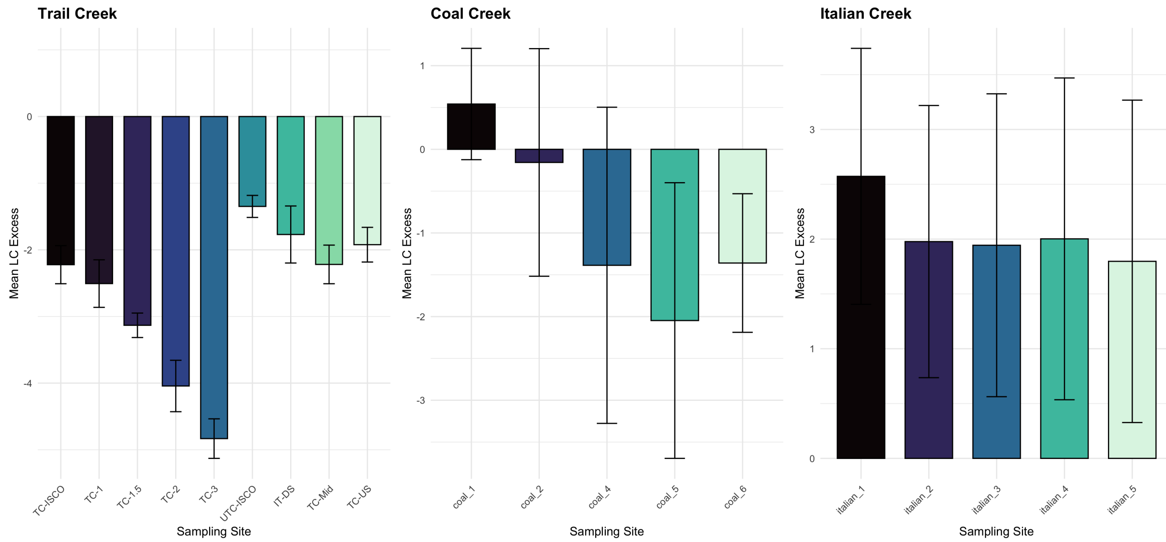
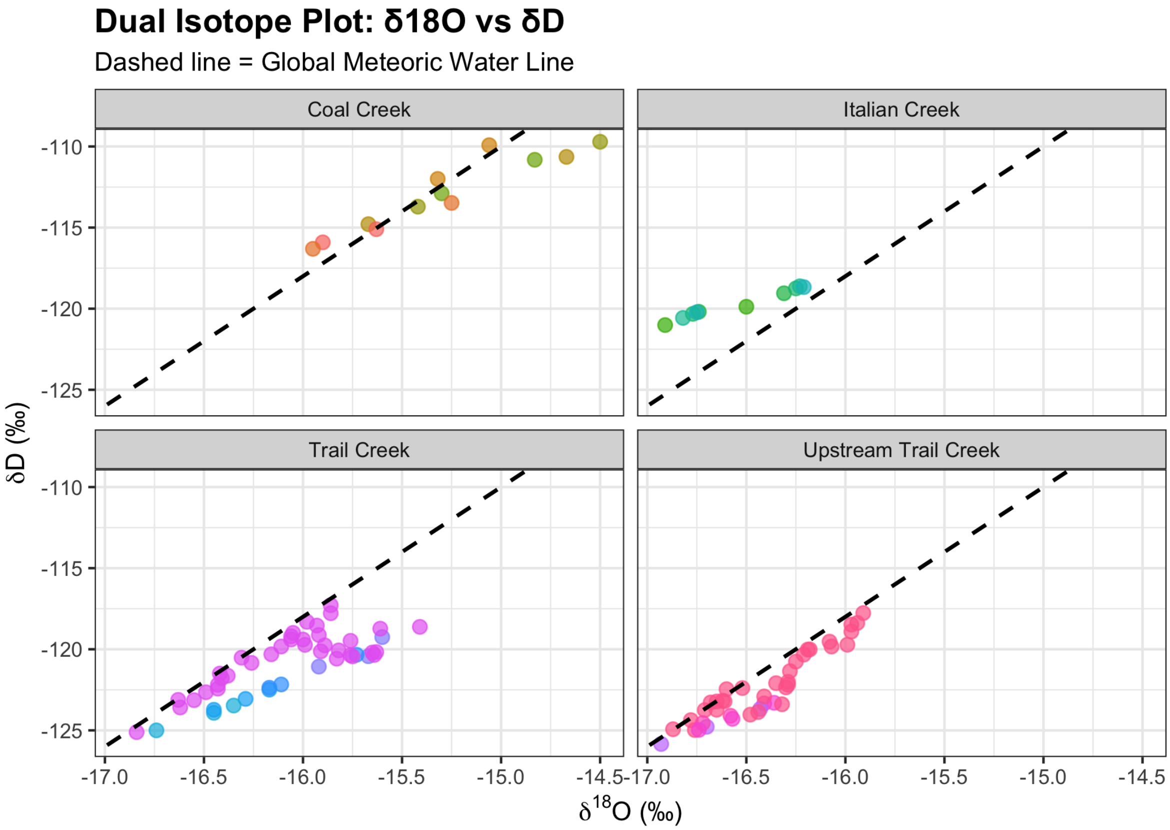


Figure 4:

```
# Add Global Meteoric Water Line (GMWL: y = 8x + 10)
gmwl_slope <- 8.00
gmwl_intercept <- 10.00

plot6 <- ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_abline(slope = gmwl_slope, intercept = gmwl_intercept,
              linetype = "dashed", color = "black", linewidth = 0.8) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         utc = "Upstream Trail Creek",
                                                         cc = "Coal Creek",
                                                         it = "Italian Creek"))) +
  labs(title = "Dual Isotope Plot: \u03b418O vs \u03b4D",
       subtitle = "Dashed line = Global Meteoric Water Line",
       x = expression(paste(delta^18, "O (\u2030)")),
       y = expression(paste(delta, "D (\u2030)")),
       color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "none")

plot6
```



Dual Isotope Plot: δ18O vs δD
Dashed line = Global Meteoric Water Line
Coal Creek
Italian Creek
Trail Creek
Upstream Trail Creek
δD (‰)
-110
-115
-120
-125
-17.0
-16.5
-16.0
-15.5
-15.0
-14.5
δ18O (‰)